AI Short Explainer Video with Voice-over

Automated Video Content Generation Pipeline

Project Overview

This project is designed to automatically generate engaging, short-form explainer videos using Artificial Intelligence. By combining AI-driven script writing, automated visual asset synthesis, natural text-to-speech voiceovers, and programmatic video rendering, the pipeline transforms text prompts or topics into polished videos ready for social media platform publishing.

Key Features

- **Automated Script & Storyboard Generation:** Leverages LLMs to craft concise, engaging scripts with structured scene-by-scene visual cues.
- **Realistic AI Voice-over:** Converts generated text scripts into human-like audio using state-of-the-art TTS models with custom voice selection.
- **Visual Asset Synthesis:** Automatically generates relevant high-resolution images or short video clips matching the narrative context.
- **Dynamic Captions & Subtitles:** Automatically syncs and overlays animated captions for enhanced engagement and viewer retention.
- **Programmatic Video Assembly:** Merges background audio, voice-overs, visual assets, transitions, and overlays into a final high-definition MP4.

Architecture & Tech Stack

Script & Narrative

OpenAI GPT-4 / Anthropic Claude / Google Gemini API

Voice-Over & Audio

ElevenLabs API / OpenAI TTS / Edge-TTS for realistic

Visuals & Assembly

Midjourney / Stable Diffusion / Runway API for visuals; MoviePy / FFmpeg /

for structured prompt processing and scriptwriting.

narration and background audio sync.

Remotion for video rendering.

Execution Workflow

Step 1 **Input Topic & Prompt Definition:** User provides a core topic, target audience, and desired tone.

Step 2 **Script Generation:** LLM outputs formatted JSON containing visual descriptions, narration text, and timing breakdowns.

Step 3 **Audio & Visual Asset Creation:** Text-to-speech generates audio clips, while image/video models generate matching visuals for each scene.

Step 4 **Assembly & Export:** Video composition engine aligns visuals with audio timings, adds dynamic captions, applies transitions, and exports the final file.